Shahjalal University of Science and Technology, Sylhet

Department of Electrical & Electronic Engineering

Course Code: EEE 0713 3237 Course Title: Power System II

Semester Final Examination (OBE): **Final Model Question**

FINAL SET 1

Practice paper for self-study. Built from previous-year questions, textbook examples/exercises and lecture notes, mapped to CO1–CO5. Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

*[Marks · CO · BL = Bloom level (C1 Remember … C5 Evaluate) · Source: **PYQ** = previous-year question, TB = textbook example/exercise (Ex = worked example, P/Tut = exercise), LN = lecture-note article. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]*

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL	Source
1.	(a)	Define self GMD (GMR) and mutual GMD. Why are bundled conductors used in EHV lines?	[2]	CO1	C1	LN 2.10
	(b)	Derive the expression for the inductance of composite-conductor lines.	[4]	CO1	C3	**PYQ**
	(c)	The forward circuit of a single-phase line is composed of three solid wires of 2.5 mm radius and the return circuit of two wires of 5 mm radius, placed symmetrically as shown in Fig. 1. Find the inductance of each side of the line and of the complete line.	[4]	CO1	C3	TB Ex 2.3

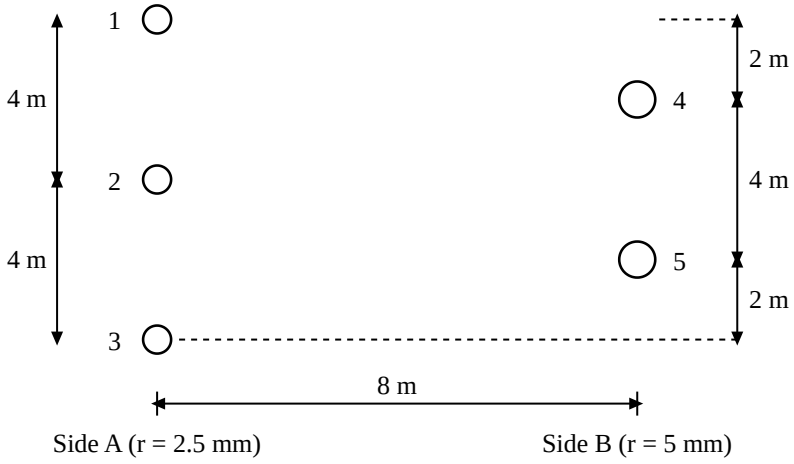


Fig. 1

2.	(a)	Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing (fully transposed).	[5]	CO1	C3	**PYQ**
	(b)	Find the capacitance to neutral per km of a three-phase line having conductors of 2 cm diameter placed at the corners of a triangle with sides 5 m, 6 m and 7 m. Also find the charging current per km if the line operates at 132 kV, 50 Hz.	[3]	CO1	C3	TB P 3.8
	(c)	How does the presence of earth affect the capacitance of a transmission line? When can it be neglected?	[2]	CO1	C2	TB 3.7

3.	(a)	With a neat figure, describe the construction of a 3-core belted cable. Why are belted cables limited to about 11 kV?	[3]	CO2	C2	LN 11.1/11.5
	(b)	Derive the expression for the insulation resistance of a single-core cable.	[3]	CO2	C3	TB 11.7
	(c)	A single-core cable has a conductor diameter of 1 cm and insulation thickness of 0.4 cm. If the specific resistance of insulation is 5×10^{14} Ω-cm, calculate the insulation resistance for a 2 km length.	[2]	CO2	C3	TB Ex 11.1
	(d)	A single-core cable 5 km long has an insulation resistance of 0.4 MΩ. The core diameter is 20 mm and the diameter of the cable over the insulation is 50 mm. Calculate the resistivity of the insulating material.	[2]	CO2	C3	TB Ex 11.3
		OR				
4.	(a)	What is capacitance grading? For three dielectrics, derive the expression for the potential difference between core and earthed sheath.	[4]	CO2	C3	LN 11.12
	(b)	A single-core lead-sheathed cable is graded using three dielectrics of relative permittivity 5, 4 and 3. The conductor diameter is 2 cm and the overall diameter is 8 cm. If the three dielectrics are worked at the same maximum stress of 40 kV/cm, find the safe working voltage. What would be the safe working voltage of an ungraded cable of the same dimensions? Comment on the result.	[4]	CO5	C5	TB Ex 11.12
	(c)	Why is intersheath grading rarely used in practice?	[2]	CO2	C2	TB 11.13

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Source
5.	(a)	Draw the equivalent circuit of a 3-disc suspension string and explain why the voltage across the discs is not equally distributed.	[2]	CO2	C2	LN 8.6
	(b)	Each conductor of a 3-phase high-voltage line is suspended by a string of 4 suspension-type discs. If the potential difference across the second unit from the top is 13.2 kV and across the third from the top is 18 kV, determine K and the voltage between conductors.	[4]	CO2	C4	TB Ex 8.8
	(c)	Each line of a 3-phase system is suspended by a string of 3 identical insulators of self-capacitance C. The shunt capacitance of the connecting metal work of each insulator is 0.2 C to earth and 0.1 C to line. Calculate the string efficiency if a guard ring increases the capacitance to the line of the metal work of the lowest insulator to 0.3 C.	[4]	CO2	C4	TB Ex 8.11
6.	(a)	Define critical disruptive voltage. Write down its significance.	[2]	CO2	C1	**PYQ**
	(b)	What factors affect corona? Why are ACSR and bundled conductors preferred for EHV lines from the corona point of view?	[2]	CO5	C5	TB 8.11/8.14
	(c)	A 3-phase, 220 kV, 50 Hz line has 1.5 cm radius conductors spaced 2 m apart in an equilateral triangle. If the temperature is 40 °C, the pressure is 76 cm Hg and $m_0 = 0.85$, calculate the corona loss per km of the line.	[4]	CO4	C3	TB Ex 8.15
	(d)	A 3-phase line has conductors 2 cm in diameter spaced equilaterally 1 m apart. If the dielectric strength of air is 30 kV (max) per cm, find the disruptive critical voltage ($\delta = 0.952$, $m_0 = 0.9$).	[2]	CO4	C3	TB Ex 8.13

OR

7.	(a)	Classify the faults in underground cables. How is a Megger used to identify each type?	[3]	CO4	C2	TB 11.20
	(b)	With a neat diagram, derive the expression for the fault distance in the Murray loop test (earth fault).	[3]	CO4	C3	LN 11.22
	(c)	In a Murray loop test for a ground fault on 500 m of cable having a resistance of 1.6 Ω/km, the faulty cable is looped with a sound cable of the same length and cross-section. If the ratio of the other two arms at balance is 3 : 1, find the distance of the fault from the testing end.	[2]	CO4	C3	TB Ex 11.22
	(d)	A Varley loop test is performed to locate an earth fault on a 20 km long cable. The resistance per km of the single conductor is 20 Ω. The loop is completed with a similar healthy conductor. At balance, the variable resistance connected to the faulty conductor is 200 Ω and the fixed resistors are equal. Calculate the distance of the fault from the test end.	[2]	CO4	C3	**PYQ**
8.	(a)	Derive the swing equation for a synchronous machine.	[4]	CO4	C3	**PYQ**
	(b)	A 50 Hz, four-pole turbogenerator rated 100 MVA, 11 kV has an inertia constant of 8.0 MJ/MVA. (i) Find the stored energy at synchronous speed. (ii) If the mechanical input is suddenly raised to 80 MW for an electrical load of 50 MW, find the rotor acceleration. (iii) If this acceleration is maintained for 10 cycles, find the change in torque angle and the rotor speed (rpm) at the end of this period.	[4]	CO4	C3	TB Ex 12.1
	(c)	Define the steady-state stability limit. How do series capacitors and FACTS devices raise it?	[2]	CO3	C2	TB 12.6.1 / LN

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FINAL SET 2

Practice paper for self-study. Built from previous-year questions, textbook examples/exercises and lecture notes, mapped to CO1–CO5. Not an official question paper.

Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

*[Marks · CO · BL = Bloom level (C1 Remember … C5 Evaluate) · Source: **PYQ** = previous-year question, TB = textbook example/exercise (Ex = worked example, P/Tut = exercise), LN = lecture-note article. Answer Q1, Q2 and either Q3 or Q4 from Part A; Q5, Q8 and either Q6 or Q7 from Part B.]*

Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL	Source
1.	(a)	Define inductance. Show that the internal inductance of a conductor is ½ × 10^{-7} H/m.	[3]	CO1	C3	TB 2.2/2.3
	(b)	The outside diameter of the single layer of aluminium strands of an ACSR conductor (6 Al + 1 steel, all strands the same size) is 5.04 cm, and the diameter of each strand is 1.68 cm. Determine the 50 Hz loop reactance at 1 m spacing. Neglect the central steel strand and give the reason.	[3]	CO1	C3	TB Ex 2.2
	(c)	Find the inductive reactance in Ω/km at 50 Hz of the three-phase bundled-conductor line of Fig. 1 (two conductors per phase, ACSR radius 1.725 cm, s = 0.4 m, d = 7 m). Compare it with an equivalent single-conductor line of the same total area.	[4]	CO1	C4	TB Ex 2.7

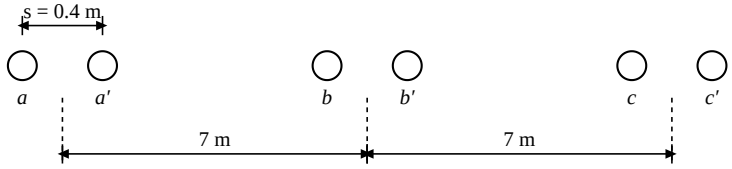


Fig. 1

			Marks	CO	BL	Source
2.	(a)	What is transposition of a transmission line? Why is it used?	[2]	CO1	C2	**PYQ**
	(b)	Derive the inductance per phase of a three-phase line with unsymmetrical spacing, assuming full transposition.	[4]	CO1	C3	LN 2.8
	(c)	A three-phase line with equilateral spacing of 3 m is to be rebuilt with horizontal spacing ($D_{13} = 2D_{12} = 2D_{23}$), fully transposed. Find the spacing between adjacent conductors such that the new line has the same inductance as the original line.	[2]	CO1	C3	TB P 2.13
	(d)	Calculate the capacitance to neutral per km of a single-phase line of No. 2 single-strand conductors (radius 0.328 cm) spaced 3 m apart and 7.5 m above the ground, (i) neglecting and (ii) considering the effect of earth.	[2]	CO1	C3	TB Ex 3.1

3.	(a)	Derive the expression for dielectric stress in a single-core cable and prove that it is maximum at the conductor surface.	[3]	CO2	C3	**PYQ**
	(b)	A single-core lead-sheathed cable has a conductor diameter of 3 cm; the diameter of the cable is 9 cm. It is graded using two dielectrics of relative permittivity 5 and 4 with safe working stresses of 30 kV/cm and 20 kV/cm. Calculate the radial thickness of each insulation and the safe working voltage of the cable.	[4]	CO2	C3	TB Ex 11.13
	(c)	Find the economic size of a single-core cable working on a 220 kV, 3-phase system. The maximum permissible stress in the dielectric is not to exceed 250 kV/cm.	[3]	CO2	C3	TB Tut 11.10-2

OR

4.	(a)	Explain the core-to-core (C_c) and core-to-earth (C_e) capacitances of a 3-core cable. Show that $C_N = C_e + 3C_c$ and explain how C_e and C_c are measured.	[3]	CO2	C2	LN 11.14/11.15
	(b)	The capacitances of a 3-phase belted cable are 12.6 µF between the three cores bunched together and the lead sheath, and 7.4 µF between one core and the other two connected to the sheath. Find the charging current drawn by the cable when connected to 66 kV, 50 Hz supply.	[3]	CO2	C3	**PYQ**
	(c)	The capacitance per km of a 3-phase belted cable is 0.18 µF between two cores with the third core connected to the sheath. Calculate the kVA taken by a 20 km long cable when connected to a 3-phase, 50 Hz, 3300 V supply.	[4]	CO2	C3	TB Ex 11.19

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Source
5.	(a)	Derive the expressions for sag when (i) the supports are at equal levels and (ii) the supports are at unequal levels.	[4]	CO2	C3	LN 8.16
	(b)	A transmission line has a span of 200 m between level supports. The conductor has a cross-sectional area of 1.29 cm², weighs 1170 kg/km and has a breaking stress of 4218 kg/cm². Calculate the sag for a safety factor of 5, allowing a wind pressure of 122 kg per m² of projected area. What is the vertical sag?	[3]	CO2	C3	TB Ex 8.19
	(c)	A 132 kV transmission line has: weight of conductor = 680 kg/km, span = 260 m, ultimate strength = 3100 kg, safety factor = 2. Calculate the height above ground at which the conductor should be supported. Ground clearance required is 10 m.	[3]	CO2	C3	**PYQ**

6.	(a)	Name the types of insulators with their voltage ranges. Select and justify a suitable type for (i) an 11 kV rural feeder and (ii) a 230 kV line crossing a river.	[2]	CO5	C5	LN 8.5
	(b)	A string of four insulators has a self-capacitance equal to 5 times the pin-to-earth capacitance. Find (i) the voltage distribution across the units as a percentage of the total voltage across the string and (ii) the string efficiency.	[4]	CO2	C4	TB Ex 8.9
	(c)	It is required to grade a string having seven suspension insulators. If the pin-to-earth capacitances are all equal to C, determine the line-to-pin capacitance that would give the same voltage across each insulator of the string.	[4]	CO2	C4	TB Ex 8.12

OR

7.	(a)	What is corona? Write down its advantages and disadvantages.	[3]	CO4	C2	**PYQ**
	(b)	Taking the dielectric strength of air as 30 kV/cm, calculate the disruptive critical voltage for a 3-phase line with conductors of 1 cm radius spaced symmetrically 4 m apart.	[3]	CO4	C3	TB Tut 8.14-2
	(c)	A certain 3-phase equilateral transmission line has a total corona loss of 53 kW at 106 kV and a loss of 98 kW at 110.9 kV. What is the disruptive critical voltage? What is the corona loss at 113 kV?	[4]	CO4	C4	TB Ex 8.16
8.	(a)	Using the P–δ curve, explain what happens for a sudden increase in mechanical input to a generator.	[3]	CO4	C2	**PYQ**
	(b)	For the system of Fig. 2, the generator delivers 1.0 pu with $E' = 1.075\angle 33.9°$ pu, and P_{max} during a 3-phase fault at the middle of one line is 0.694 pu. If H = 4 MJ/MVA, write the swing equation upon occurrence of the fault. Find the initial angular acceleration and, assuming it remains constant for $\Delta t = 0.05$ s, the rotor angle at the end of this interval and the new acceleration.	[4]	CO4	C4	TB Ex 12.4

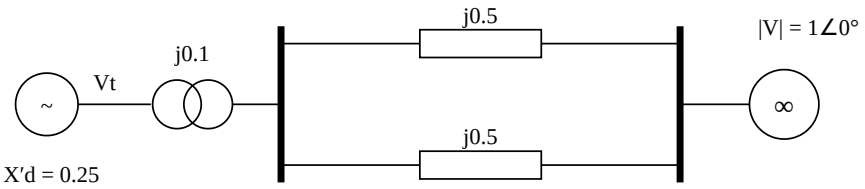


Fig. 2

	(c)	Find the steady-state power limit of a system consisting of a generator of equivalent reactance 0.50 pu connected to an infinite bus through a series reactance of 1.0 pu, when the terminal voltage of the generator is held at 1.20 pu and the infinite-bus voltage is 1.0 pu. Compare it with the limit when the generator emf is held fixed at 1.2 pu, and comment on the role of the voltage regulator.	[3]	CO3	C4	TB Ex 12.6

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FINAL SET 3

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Total Marks: 60 **Credits: 3.0** **Time: 2 Hours (approx.)**

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Part A

Answer Q1, Q2 and EITHER Q3 OR Q4 (3 × 10 = 30)

			Marks	CO	BL	Source
1.	(a)	Derive the expression for the flux linkages of one conductor in a group of conductors.	[4]	CO1	C3	**PYQ**
	(b)	A double-circuit three-phase line is shown in Fig. 1. Conductors a, a′; b, b′; c, c′ belong to the same phase respectively. The radius of each conductor is 1.5 cm. Find the inductance of the double-circuit line in mH/km/phase (assume transposition).	[3]	CO1	C3	TB P 2.12

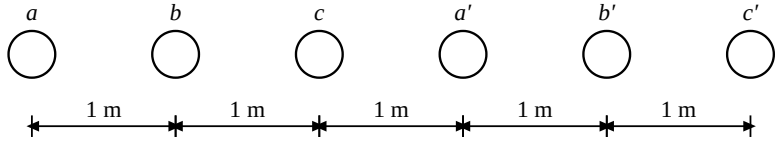


Fig. 1

	(c)	A telephone line runs parallel to an untransposed three-phase line as shown in Fig. 2. The power line carries a balanced current of 400 A per phase. Find the mutual inductance between the circuits and calculate the 50 Hz voltage induced in the telephone line per km.	[3]	CO1	C4	TB P 2.8

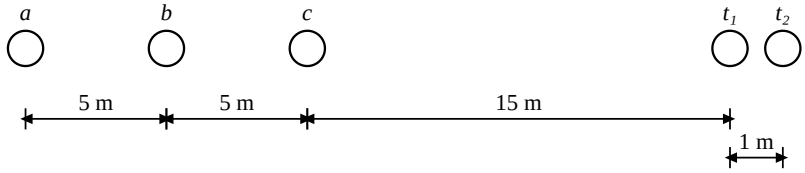


Fig. 2

2.	(a)	Define skin effect and proximity effect. Why is the effective AC resistance higher than the DC resistance?	[2]	CO1	C2	**PYQ**
	(b)	Derive the capacitance of a single-phase two-wire line and show that the capacitance to neutral is twice the line-to-line capacitance.	[3]	CO1	C3	TB 3.4
	(c)	A three-phase, 50 Hz overhead line has regularly transposed conductors equilaterally spaced 4 m apart. The capacitance of such a line is 0.01 μF/km. Recalculate the capacitance per km to neutral when the conductors are in the same horizontal plane with successive spacing of 4 m and are regularly transposed.	[2]	CO1	C3	TB P 3.3
	(d)	Consider the 500 kV, three-phase bundled-conductor line of Fig. 3 (2 conductors per phase, diameter 30 mm). Find the capacitive reactance to neutral in Ω-km at 50 Hz.	[3]	CO1	C3	TB P 3.4

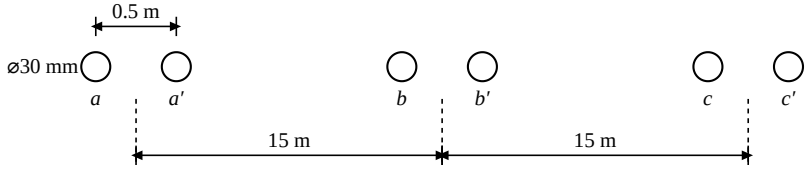


Fig. 3

3.	(a)	Compare the merits and demerits of the underground system against the overhead system. Briefly describe the methods of laying underground cables.	[3]	CO5	C5	TB 11.6 / Review
	(b)	Calculate the capacitance and charging current of a single-core cable used on a 3-phase, 66 kV system. The cable is 1 km long with a core diameter of 10 cm and an impregnated paper insulation of thickness 7 cm. The relative permittivity of the insulation is 4 and the supply is 50 Hz.	[3]	CO2	C3	**PYQ**
	(c)	A single-core cable of conductor diameter 2 cm and lead sheath of diameter 5.3 cm is to be used on a 66 kV, 3-phase system. Two intersheaths of diameter 3.1 cm and 4.2 cm are introduced between the core and lead sheath. If the maximum stress in the layers is the same, find the voltages on the intersheaths.	[4]	CO2	C4	TB Ex 11.15

OR

4.	(a)	Define thermal resistance. Derive the thermal resistance of the dielectric of a single-core cable and the expression for the permissible current loading.	[4]	CO2	C3	LN 11.17–11.19
	(b)	The capacitance per km of a 3-phase belted cable is 0.3 μF between two cores with the third core connected to the lead sheath. Calculate the charging current taken by five km of this cable when connected to a 3-phase, 50 Hz, 11 kV supply.	[3]	CO2	C3	TB Ex 11.17
	(c)	The capacitances per km of a 3-phase cable are 0.63 μF between the three cores bunched together and the sheath, and 0.37 μF between one core and the other two connected to the sheath. Calculate the charging current taken by eight km of this cable when connected to a 3-phase, 50 Hz, 6600 V supply.	[3]	CO2	C3	TB Tut 11.15-1

Part B

Answer Q5, Q8 and EITHER Q6 OR Q7 (3 × 10 = 30)

			Marks	CO	BL	Source
5.	(a)	Discuss the methods of improving string efficiency.	[3]	CO2	C2	**PYQ**
	(b)	An insulator string consists of three units, each having a safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency.	[3]	CO2	C4	TB Ex 8.5
	(c)	A string of 4 insulators has a self-capacitance equal to 10 times the pin-to-earth capacitance. Find (i) the voltage distribution across the units as a percentage of the total voltage across the string and (ii) the string efficiency.	[4]	CO2	C4	TB Ex 8.6
6.	(a)	Explain the effect of wind and ice loading on sag. Write the expressions for slant sag and vertical sag.	[3]	CO2	C2	LN 8.16
	(b)	An overhead conductor with a parabolic configuration weighs 1.925 kg per metre. The area of cross-section is 2.2 cm^2 and the ultimate strength is 8000 kg/cm^2. The supports are 600 m apart with a 15 m difference of levels. Calculate the sag from the taller of the two supports for a factor of safety of 5. Assume an ice load of 1 kg per metre run and no wind pressure.	[4]	CO2	C3	TB Ex 8.24
	(c)	An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2, ultimate strength 5000 kg/cm^2, specific gravity 8.9 gm/cc and safety factor 5. The wind pressure is 1.5 kg/m. Calculate the height of the conductor above the ground at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor.	[3]	CO2	C3	TB Ex 8.22
		OR				
7.	(a)	With a neat diagram, explain the Varley loop test for locating an earth fault and derive the fault-distance expression.	[3]	CO4	C3	LN 11.23
	(b)	A Murray loop test is performed on a faulty cable 300 m long. At balance, the resistance connected to the faulty core was set at 15 Ω and the resistance of the resistor connected to the sound core was 45 Ω. Calculate the distance of the fault from the test end.	[2]	CO4	C3	TB Ex 11.21
	(c)	The Varley loop test is used to find the position of an earth fault on a line of length 40 km. The resistance per km of a single line is 28 Ω. The fixed resistors are 250 Ω each. The fault is calculated to be 7 km from the test end. To what value was the variable resistor set?	[3]	CO4	C4	TB Tut 11.23-3
	(d)	In a test for a fault to earth on a 500 m length of cable having a resistance of 1 Ω per 1000 m, the faulty cable is looped with a sound cable of the same length but 2.25 Ω per 1000 m. The other two arms at balance are in the ratio 2.75 : 1. Calculate the distance of the fault.	[2]	CO4	C3	**PYQ**

8.	(a)	Define HVDC. Discuss the economics of power transmission with a cost–distance curve.	[3]	CO5	C2	**PYQ**
	(b)	Explain series and shunt compensation of transmission lines. Name the main FACTS controllers and their applications.	[3]	CO3	C2	LN 16/17
	(c)	A generator (X_{dg} = 1 pu) feeds a motor (X_{dm} = 1 pu) through a transformer (0.1 pu), a line (0.25 pu) and a transformer (0.1 pu). A 3-phase static capacitive reactor of 1 pu per phase can be connected through a switch at the motor-side bus (between the line and the motor transformer). Taking E_g = 1.2 pu and E_m = 1.0 pu, calculate the steady-state power limit with and without the reactor. Recalculate when the capacitive reactor is replaced by an inductive reactor of the same value, and comment.	[4]	CO4	C4	TB Ex 12.2

How These Sets Were Built & Answer Key

Why this mix

You told me the course teacher (i) follows the course learning outcomes, (ii) reuses about 20 % from the last 4 years' finals and (iii) takes the rest from the textbook chapters and lecture notes. The external examiner can then edit the draft. So each set has:

- **PYQ (about 20 %)**: the repeats most likely to come back, word-for-word with the same numbers.
- **TB (about 50–60 %)**: worked examples and end-of-chapter exercises from Kothari Ch-2, 3, 12 and Mehta Ch-8, 11 (the books in 01_Textbook_Chapters). Teachers lift these almost unchanged, and the external examiner usually swaps in another example of the same type.
- **LN (about 20–25 %)**: derivations and theory articles the teacher covered in the lectures (02_Lecture_Notes).
- **CO wording**: "compare / justify / comment" endings on some parts for CO2, CO3 and CO5, which is how an external examiner usually modifies an OBE paper.

Marks per CO (80 printed marks per set) and source share of the 60 answered marks

Set	CO1	CO2	CO3	CO4	CO5	PYQ	Textbook	Lecture notes
Final Set 1	20	28	2	24	6	15 (25 %)	34–38 (57–63 %)	7–11 (12–18 %)
Final Set 2	20	38	3	17	2	11–14 (18–23 %)	35–39 (58–65 %)	8–13 (13–22 %)
Final Set 3	20	37	3	14	6	12–17 (20–28 %)	36–39 (60–65 %)	6–10 (10–17 %)

Ranges depend on which optional questions (Q3/Q4, Q6/Q7) are answered.

Final Set 1: final answers (textbook answers re-checked)

- 1(c) D_m = 8.8 m, D_{sA} = 0.367 m, D_{sB} = 0.125 m; L_A = 0.635, L_B = 0.85, L = 1.485 mH/km
- 2(b) D_{eq} = 5.94 m; C_n = 0.00872 µF/km; I_c = 0.209 A/km
- 3(c) 234 MΩ · 3(d) ρ = 13.72 × 10^9 Ω-m
- 4(b) d_1 = 2.5 cm, d_2 = 3.34 cm; graded 57.8 kV vs ungraded 39.2 kV (rms), about 47 % higher
- 5(b) K = 0.2; V_1 = 11, V_4 = 26.49 kV; V_{ph} = 68.69 kV → V_L = 119 kV · 5(c) η = 95 %
- 6(c) δ = 0.952; V_c = 125.9 kV/ph; P = 0.06 kW/km (3 phases) · 6(d) V_c = 83.64 kV/ph → 144.8 kV line
- 7(c) 250 m · 7(d) 15 km [PYQ]
- 8(b) 800 MJ; α = 337.5 elec °/s²; Δδ = 6.75°; N = 1505.6 rpm

Final Set 2: final answers (textbook answers re-checked)

- 1(b) D_s = 1.155d = 1.93 cm; L = 0.789 mH/km per conductor; loop X = 0.495 Ω/km
- 1(c) D_{eq} = 8.81 m, D_s = 0.073 m; X = 0.301 Ω/km (single equivalent conductor 0.531 Ω/km, 76 % higher)
- 2(c) D = $3/\sqrt[3]{2}$ = 2.38 m · 2(d) 0.00817 µF/km (earth neglected), 0.0082 µF/km (with earth)
- 3(b) d_1 = 5.625 cm; thicknesses 1.31 cm & 1.69 cm; V = 54.7 kV peak = 38.7 kV rms · 3(c) d = 1.43 cm, D = 3.88 cm
- 4(b) 107.7 A [PYQ] · 4(c) $C_N = 2C_3$ = 7.2 µF; I_c = 4.3 A; 24.57 kVA
- 5(b) slant sag 8.96 m, vertical 5.37 m · 5(c) 13.7 m [PYQ]
- 6(b) 16, 19.2, 26.3, 38.5 %; η = 64.9 % · 6(c) C_1…C_6 = 0.167C, 0.4C, 0.75C, 1.33C, 2.5C, 6C
- 7(b) V_c = 127 kV/ph → 220 kV line · 7(c) V_c = 54 kV; W = 128 kW

- 8(b) $d^2\delta/dt^2 = 2250(1 - 0.694 \sin \delta)$; 1379 elec °/s²; $\delta = 35.6°$; new $\alpha = 1341$ elec °/s² · 8(c) 1.152 pu vs 0.8 pu

Final Set 3: final answers (textbook answers re-checked)

- 1(b) $D_s = 0.187$ m, $D_{eq} = 1.815$ m; L = 0.454 mH/km/phase
- 1(c) $|\lambda| = 1.77$ μWb-T/m; V ≈ 0.557 V/km (phase M: 0.0078, 0.0098, 0.0129 mH/km)
- 2(c) 0.0096 μF/km · 2(d) $C_n = 0.01033$ μF/km; $X_c = 3.08 \times 10^5$ Ω-km
- 3(b) 0.254 μF; 3.04 A [PYQ] · 3(c) $V_1 = 16.88$ kV; intersheaths at 37.02 kV and 18.88 kV
- 4(b) $C_N = 3$ μF; 5.98 A · 4(c) 4.31 A
- 5(b) string voltage 37.92 kV; η = 84.3 % · 5(c) 19.76, 21.74, 25.9, 32.6 %; η = 76.6 %
- 6(b) $x_1 = 270$ m, $x_2 = 330$ m; $S_2 = 45.24$ m · 6(c) slant 3.28 m, vertical 2.5 m → 9.5 m
- 7(b) 150 m · 7(c) S = 1848 Ω · 7(d) 433.3 m [PYQ]
- 8(c) 0.49 pu (no reactor); 1.244 pu (capacitive); 0.304 pu (inductive)

Study tip: the three sets together cover nearly every worked example in the five chapters. If the external examiner changes the numbers, the method is the same, so practise the method rather than memorising answers. Full PYQ solutions are in 03_Solved_Math_Bank.md; the derivation steps are in 04_Derivations_Guide.md.